the state of the object or system, and its structure—how components are organized and connected. By remaining synchronized through continuously receiving updates about current conditions, digital twins allow users to track the behavior of the physical system while also predicting the effect of changes without changing the real system.

For most of human history, you had to be physically present to understand how something worked. Digital twins have changed that, and today, countless people can simultaneously monitor, analyze, and run simulations on the same physical system from opposite sides of the globe.

1.2 Technology enabling digital twins

Digital twins first gained traction in capital-intensive industries like manufacturing, utilities, and energy. These industries require massive investments, often operate in remote or hazardous environments, and face strict safety and regulatory requirements. Since equipment costs are high, even small improvements in efficiency or reductions in downtime quickly justify technology investments.

While these industries have used operational technology (OT) systems like programmable logic controllers (PLC) and supervisory control and data acquisition (SCADA) for monitoring and control since the 1960s, digital twins represent a major leap forward. By integrating IoT sensors, cloud computing, and AI/ML, modern digital twins can process vast amounts of data, run sophisticated predictive models, and provide actionable insights that traditional control systems alone could never deliver.

The convergence of these technologies enables three key capabilities: virtual modeling during design to reduce physical prototyping, predictive maintenance to prevent expensive failures, and real-time optimization to maximize operational efficiency.

Consider a wind farm where engineers first use the digital twin to virtually test turbine designs under simulated conditions. Once deployed, IoT sensors stream data to the cloud where AI algorithms detect early signs of bearing wear. The digital twin then simulates maintenance scenarios and schedules